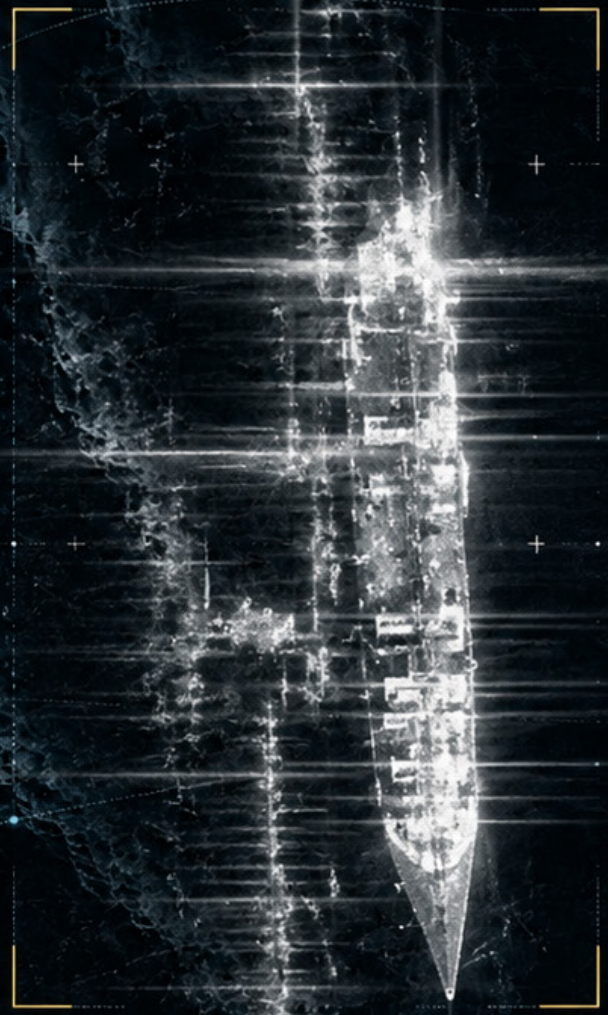
THE DARK FLEET

Revealing the vessels hidden from
public monitoring systems



A self-initiated visual communication project based
on public research and data by Global Fishing Watch

THE OCEAN WE CAN SEE

AIS has transformed the way vessel activity can be monitored at sea, but it only reveals the ships that transmit.

The Automatic Identification System (AIS) continuously broadcasts information such as a vessel's position, speed and identity.

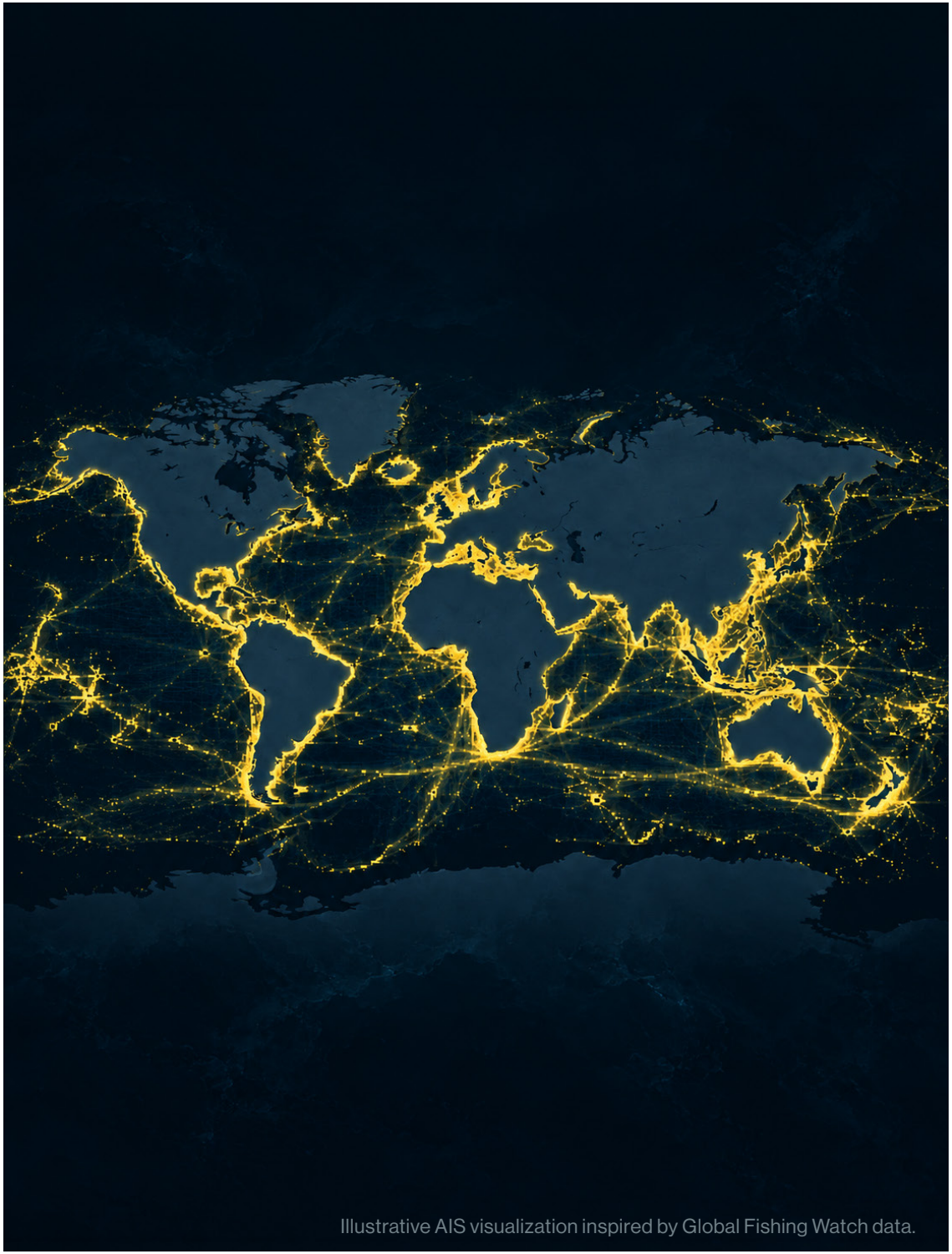
Originally developed to improve maritime safety, AIS data now allows researchers to reconstruct vessel movements across vast areas of the ocean.

But this picture is incomplete. AIS only shows vessels that are equipped with the system and **actively transmitting a signal.**

VISIBLE ≠ COMPLETE

AIS only reveals vessels that transmit.

VESSEL → AIS SIGNAL → POSITION · SPEED · IDENTITY



Illustrative AIS visualization inspired by Global Fishing Watch data.



WHAT IS A DARK VESSEL?

A vessel can be physically present at sea while remaining invisible to public tracking systems.

A vessel may not appear in public AIS data because it is not equipped with AIS, is not required to carry it, is outside receiver coverage, or is not transmitting a signal.

Satellite sensors can reveal vessels that AIS does not show, helping researchers compare what is being transmitted with what is actually present at sea.

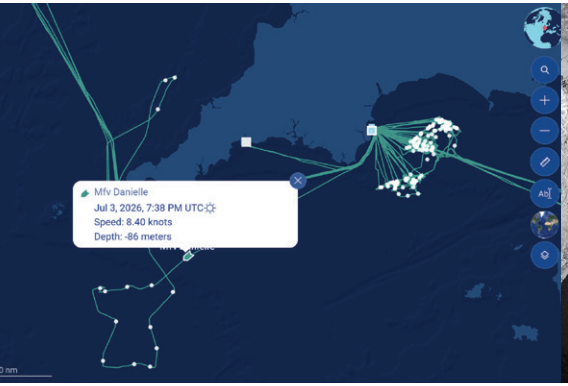
DARK ≠ ILLEGAL

A missing AIS signal alone does not prove illegal activity.

SEEING BEYOND AIS

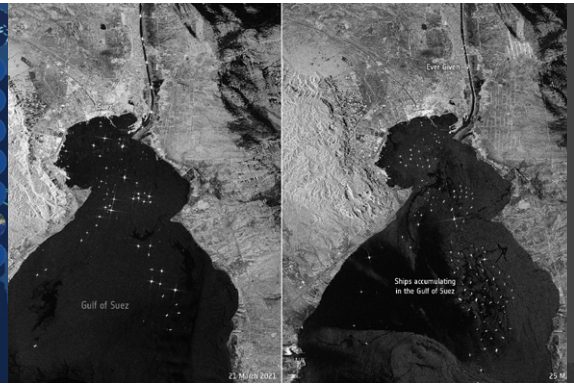
Different sensing technologies reveal different aspects of vessel activity at sea.

AIS
WHAT VESSELS TRANSMIT



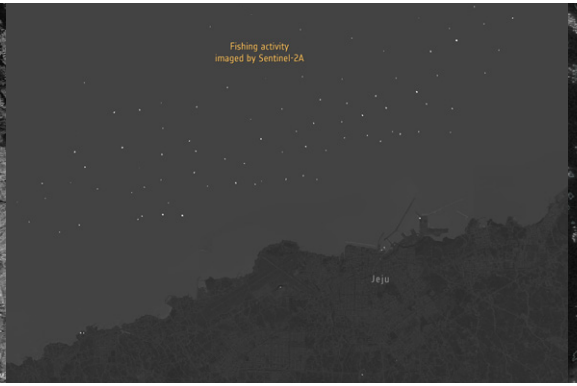
Identity, position, speed and movement are broadcast through AIS, creating a track that can be followed over time.

SAR · SENTINEL-1
WHAT RADAR CAN DETECT



SAR detects vessels from reflected microwave signals, working day or night and through most weather conditions.

OPTICAL · SENTINEL-2
WHAT SATELLITES CAN SEE



Optical imagery can reveal vessels directly at the ocean surface, including their shape, position and sometimes their wake.

VIIRS · NIGHT LIGHTS
WHAT LIGHT CAN REVEAL



VIIRS can detect bright lights associated with vessels at night, revealing activity that may not appear in public tracking data.

AIS + SAR + OPTICAL + VIIRS → A MORE COMPLETE PICTURE

Combining multiple sources helps reveal vessel activity that any single system may miss.

WHEN THE OCEAN LIGHTS UP

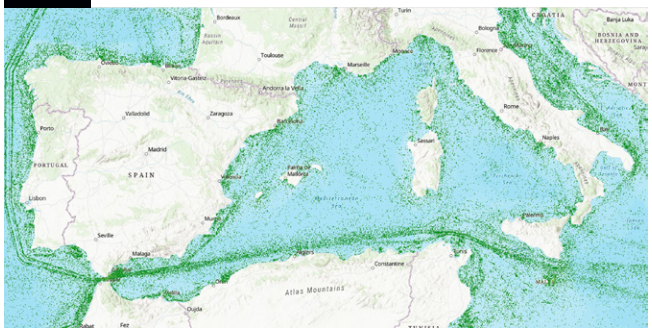
Satellite radar reveals vessel activity that public tracking alone can miss.

1 AIS APPARENT FISHING EFFORT
AIS-BASED FISHING ACTIVITY



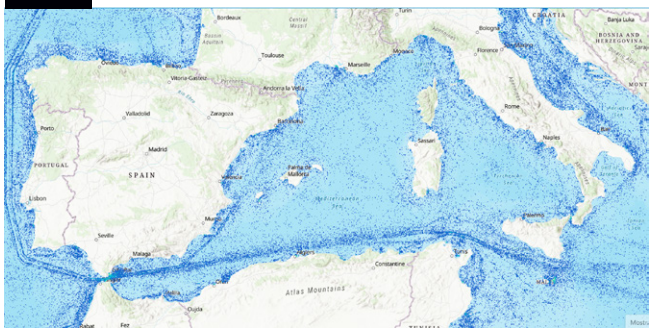
Fishing activity inferred from AIS transmissions.

2 SAR MATCHED
WHAT RADAR CONFIRMS



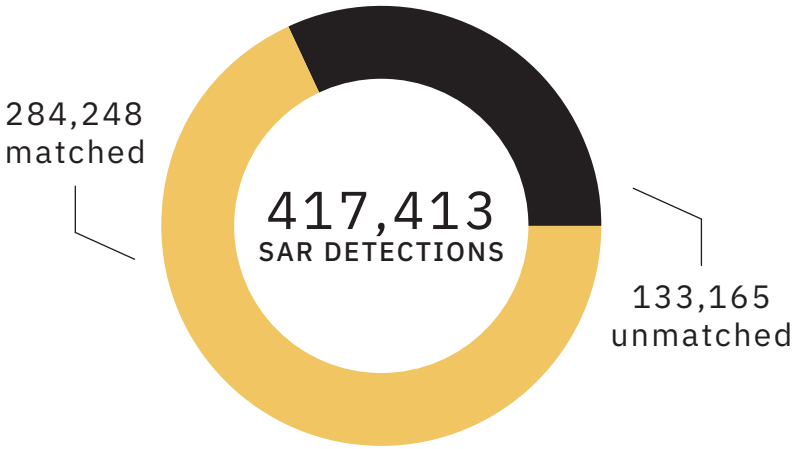
SAR detections matched to AIS-tracked vessels.

3 SAR UNMATCHED
WHAT AIS DOES NOT REVEAL



SAR detections with no AIS match.

SAR DETECTIONS · 2021



31.9% OF SAR DETECTIONS WERE UNMATCHED

↓

OF THOSE UNMATCHED DETECTIONS

71.1% CLASSIFIED AS LIKELY FISHING

UNMATCHED ≠ ILLEGAL

Area shown: 10.21°W–19.74°E, 34.12°N–45.20°N · 2021. Derived from Global Fishing Watch Sentinel-1 SAR vessel detection data.
Likely fishing classification based on GFW **fishing_score ≥ 0.9.**

WHY DARK VESSELS MATTER

Gaps in public tracking can obscure the true scale and distribution of human activity at sea.

FISHERIES MANAGEMENT

Incomplete visibility can distort our understanding of where and how fishing activity takes place.

MARINE PROTECTION

Untracked vessel activity can make it harder to monitor protected or sensitive areas.

COMPLIANCE & ENFORCEMENT

Limited visibility can make suspicious or unauthorized activity harder to investigate.

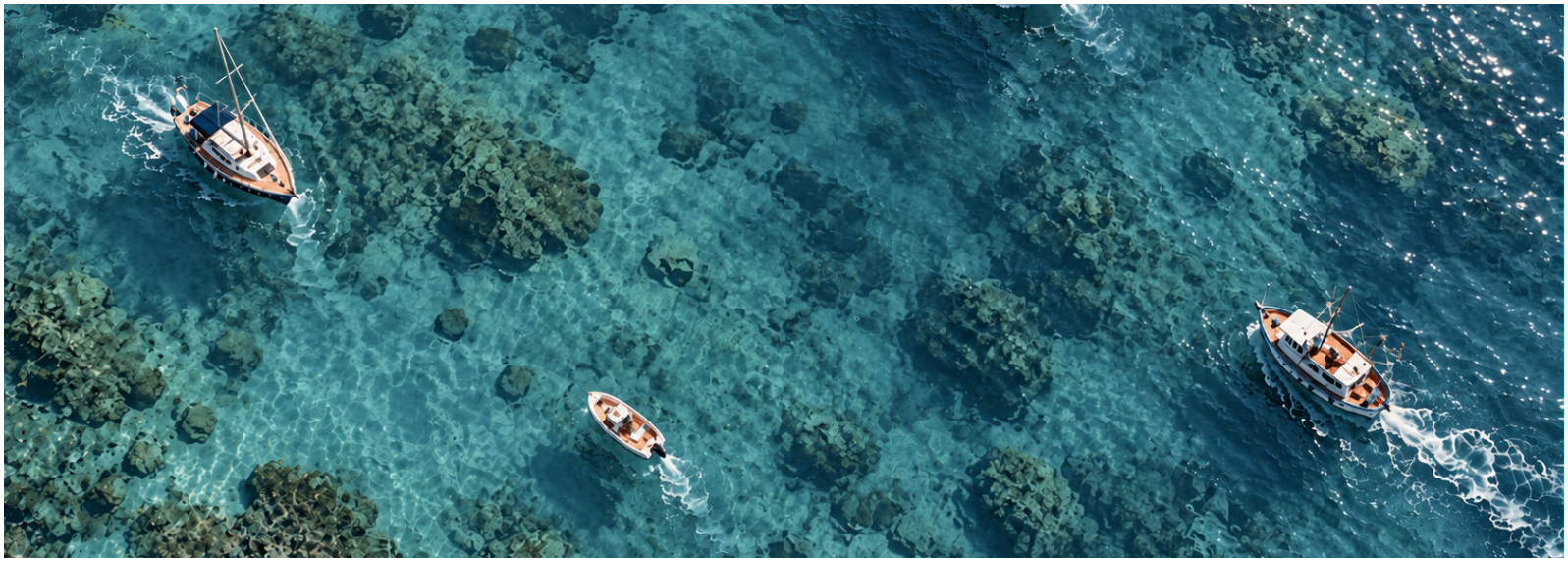
RESEARCH & TRANSPARENCY

Better detection builds a more complete picture of human activity at sea.

BETTER VISIBILITY → BETTER DECISIONS



A MORE VISIBLE OCEAN



AIS alone cannot reveal the full picture.
SAR, optical imagery and night-light detection help fill the gaps, supporting better research, stronger monitoring and **greater transparency across the ocean.**

WHAT WE CAN SEE, WE CAN BETTER UNDERSTAND

GLOBAL FISHING WATCH · SENTINEL-1 SAR · SENTINEL-2 · VIIRS

ABOUT THE RESEARCH

Based on public research and data by Global Fishing Watch on vessel activity, AIS tracking and satellite-based detection, including Sentinel-1 SAR vessel detections and research on activity that remains unmatched to public tracking systems.

READ THE SOURCE RESEARCH

This visual summary highlights selected concepts and findings and does not replace the original research. Public data has been selected, filtered and reorganised for visual communication purposes. Derived figures are based on Global Fishing Watch datasets for the geographic area and period indicated in the document.

Visual summary, data interpretation & design by María Vera.
The project combines publicly available Global Fishing Watch data and source imagery with original and AI-generated illustrative visuals.

Independent concept project. Not commissioned by or affiliated with Global Fishing Watch.